TECHNICAL PROGRESS REPORT

Monolithic rigid-body motion in beamFoam

Development history, code architecture, verification cases, partitioned-versus-monolithic comparisons, and present validation status

REPOSITORY	beamFoam, branch `colm_6DOF_solver`
EVIDENCE DATE	13 October 2026; tests recorded through 12 October
REVISION	`997a566`, including welded coupling and twist tests
SCOPE	Rigid body attached to a geometrically exact beam; standalone beamFoam tests and moorFV/interFoam coupling
PREPARED FROM	Source code, commit history, validation scripts, case READMEs, summaries, and completed run outputs

Repository evidence
Committed code + local run outputs

EXECUTIVE SUMMARY

What has been achieved

The true monolithic beam-body solver is implemented, analytically verified for translation and rotation, integrated with moorFV, and shown to be more robust than partitioned fixed-point coupling when the beam/body stiffness ratio becomes severe.

54/54

standalone analytic, Jacobian, and cross-method checks passed after rotational coupling

2.2e-10

moorFV-stored versus beamFoam-solved body state across 56 current write times

2-4x

typical reduction in beam Newton work in validated standalone comparisons

Current status by capability

VALIDATED	Translation and finite-rotation residuals, Newmark state recovery, pin attachment, finite-difference body-column Jacobians, analytic dynamics, and converged partitioned agreement.
VALIDATED	Single-line moorFV integration through `beamFoamCoupled`, including fluid-loaded interFoam cases, constraints, non-zero reference placement, gravity, contact, and long 10 s wave runs.
TESTED	Welded attachment with direct end-moment transfer. Standalone twist sweeps and a moorFV torsion case have completed; the implementation and tests are committed in 99f35f7 and ffa7ca3.
MIXED RESULT	The new 20%-slack line: monolithic and MoorDyn runs finish to 20 s; the partitioned beamFoam run fails at 12.442 s during a beam Newton solve. This is strong robustness evidence, but not a complete same-duration comparison.
OPEN	Several monolithic mooring lines, static condensation, parallel attachment handling, and displaced-state restart remain outside the demonstrated capability.

Central finding. The monolithic method removes the beam/body splitting iteration: it keeps the off-diagonal coupling Jacobians in one Newton system. The strongest evidence is the stiffness sweeps, where partitioned iterations rise and then diverge while monolithic iterations remain nearly constant.

Primary evidence: `documentation/source/monolithicCouplingPlan.html`; `tutorials/rigidBodyAndBeam/rigidBodyEnd/README.md`; `tutorials/rigidBodyEndWhirl/summary.txt`; current `tutorials/rigidBodyEndTwist/summary.txt`.

SCOPE AND EVIDENCE

How to read this report

Snapshot

The inspected baseline is branch `colm_6DOF_solver` at `997a566`, including welded-joint source (`99f35f7`) and twist tests (`ffa7ca3`) committed on 12 October. Generated float-case outputs remain local and largely untracked.

Results are labelled as committed, current-output, historical, or planned. No long CFD case was rerun for this report; existing logs and post-processing files were inspected.

Evidence hierarchy

1. Solver source and configuration actually present.
2. Completed logs and numerical summary files.
3. Automated checker scripts and their recorded pass results.
4. Commit messages and development notes.
5. Plans or expected results, clearly marked as such.

Terminology

Correct prose: partitioned.

Literal legacy path: `floatRigidBody_partioned`. The directory typo is preserved only when citing files.

“Monolithic” boundary: the beam plus attached rigid body are one nonlinear system. The fluid is still coupled through PIMPLE outer correctors; this is not a fully monolithic fluid-structure solve.

Joint types: pin coupling passes force. The newly committed welded joint also transfers cross-section rotation and end moment.

Three code generations must not be conflated

Generation	Meaning	Report treatment
Classic moorFV loop	`FvBeamNewmark` and `finiteVolumeBeam` alternate body and beam solves.	PARTITIONED REFERENCE
Legacy BlockEigen switches	Force/kinematic switches appended body values and could make BlockEigen state authoritative, but the staged body rows were not the final coupled residual.	INTERMEDIATE
`rigidBodyEnd` path	Body force/moment residuals and beam-body Jacobians are assembled with the beam cell equations in one Newton solve.	TRUE MONOLITHIC

> Several May-August notes call the authoritative BlockEigen path “coupled” or “monolithic.” The September audit corrected that interpretation. Those early tests remain useful for integration and Newmark diagnostics, but are not counted as final monolithic validation.

Revision evidence: `git status`, `git log`, commits `f84a03a`, `a2dcb75`, `359e651`, `03bbd7d`, `83acbbb`.

CONCEPTS

Partitioned and monolithic coupling

PARTITIONED

Separate subsystem solves

The body is advanced from the latest beam reaction. Its attachment motion is prescribed on the beam. beamFoam then converges its own Newton solve and returns an updated end force (and, for a weld, moment). This exchange repeats to a coupling tolerance or through PIMPLE outer correctors.

Body Newmark
Use lagged beam load
Update x, Q

→

Beam Newton
Prescribe end motion
Return -Q, -M

Numerical consequence: the diagonal subsystems may be solved accurately, but the off-diagonal interaction is lagged and converged by fixed-point iteration.

MONOLITHIC

One coupled Newton system

Beam cell increments and six rigid-body increments are unknowns in the same BlockEigen matrix. Attachment compatibility is eliminated into coupling blocks. Body force/moment balances are residual rows. One linear solve returns a joint Newton correction.

Assemble
Beam + body residual
Full coupling Jacobian

→

One solve
ΔW, ΔΘ, Δx, Δθ
Update together

Numerical consequence: Newton sees the instantaneous two-way reaction, so stiffness or low mass does not create the same fixed-point stability limit.

> **Partitioned coupling lags the interaction and iterates it outside the beam solve; monolithic coupling retains the interaction inside the Newton Jacobian.**

Important system boundary

In moorFV, `beamFoamCoupled` solves the mooring line and rigid body monolithically. Hydrodynamic force, weight, and other restraints are supplied as external loads for that call. A new PIMPLE outer corrector is still required to refresh fluid/body coupling. Thus beam-body splitting is removed, but fluid added-mass splitting is not.

Code: `.../coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C:292–461`; `../sDoFRGBFvBeam/sixDoFFvBeamSolvers/{FvBeamNewmark,beamFoamCoupled}/*.C`.

CODE ARCHITECTURE

What differs in the matrix and call flow

Both methods use the same geometrically exact beam model. The difference is where the rigid-body degrees of freedom and attachment reaction are resolved.

	Beam increments $\Delta W, \Delta \Theta$ (6N)	Body increments $\Delta x, \Delta \theta$ (6)
Beam rows	Existing cell-centred beam tangent K_{beam}	**Attachment compatibility and rotation columns $K_{beam\text{-}body}$**
Body rows	**Reaction derivatives $K_{body\text{-}beam}$**	Newmark inertia, damping, gyro and geometric terms K_{body}

Concern	Partitioned	Monolithic
Body update	`newmarkDisplacement` / `newmarkRotation` from current reaction.	Force and moment residual rows; state recovered from converged x and θ.
Attachment motion	Applied as W/Theta boundary values before each full beam solve.	Current body iterate sets the boundary; its incremental dependence is carried by body columns.
Reaction	Read as `-Q` and optional `-M`, then used on next exchange.	Action and reaction use the same discrete boundary terms in the same Jacobian.
Convergence	Beam Newton convergence plus outer displacement/rotation coupling tolerance.	One residual and step test includes active body rows and increments.
moorFV solver	`FvBeamNewmark`; may use acceleration relaxation.	`beamFoamCoupled`; copies accepted beamFoam body state without a second Newmark update.

Unknown count. BlockEigen grows from 6N to 6(N+1). If body rotation is disabled, the final three rows remain inactive identity rows.

Consistency check. Central finite differences compare the six body Jacobian columns against residual perturbations for the first N steps.

Code: `RigidBodyCouplingData.H:122-190`; `BlockEigenSolverOF.C:265-467, 852-969, 1048-1066`; `coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C:595-739`.

ALGORITHM DETAIL

One time step in each method

Partitioned fixed-point exchange

1. Store the accepted body state at time n.
2. Predict body translation and rotation with Newmark using the latest beam force/moment.
3. Measure the change; optionally relax the body update.
4. Prescribe beam-end W and, for a welded joint, Theta.
5. Run a complete beam Newton solve.
6. Extract `−Q` and optional `−M`.
7. Repeat up to `nCouplingIterations` until `couplingTolerance`.

> One pass is not a converged partitioned method. In axial oscillation it grows about 15% per period; the validation reference therefore uses 2-6 exchanges per step.

Monolithic Newton step

1. Build Newmark predictors from the accepted state.
2. Set the attachment from the current body iterate.
3. Assemble beam tangent plus beam-body and body-beam blocks.
4. Assemble body force/moment residuals, inertia, damping, gyro, arm geometry and constraints.
5. Solve for ΔW, ΔTheta, Δx and Δtheta together.
6. Update beam and body iterates; include body increments in the step norm.
7. After convergence, recover velocity/acceleration and accept the state.

> Because the body residual is in force/moment units and the reaction comes from the beam boundary discretisation, the converged result should match a fully converged partitioned exchange.

Rigid-body residuals represented in code

Translation: $m/(\beta\Delta t^2)\ (x - x_{predictor}) + c\ v - F_{external} - F_{beam} = 0$

Rotation: $J\ \alpha + \omega \times (J\omega) + c_r\ \omega - M_{external} - r \times F_{beam} - M_{beam} = 0$

Orientation: $Q = \exp(\theta)\ Q_n$; $\text{attachment} = x + [Q(\theta) - I]\ r_0$

Nuance: active body residuals and increments participate in convergence, but the current X-norm denominator is formed from beam-cell W and Theta only. This has not prevented the reported validations, but is worth retaining as a review item.

Code: `rigidBodyEnd.C:430–561, 611–638`; `coupledTotalLagNewtonRaphsonBeamEvolve.C:993–1045, 1345–1443`.

DEVELOPMENT HISTORY I

From matrix prototype to a corrected formulation

9-23 FEBRUARY 2026
Prototype. Six appended unknowns, an identity 6x6 block, I/O vector, basic six-DOF equations and first coupling experiments (`ebe8931` through `55580f3`).

10 APRIL
Formal data path. Added rigid-body coupling data/API and BlockEigen wiring (`6c25265`).

30 APRIL - 1 MAY
First interFoam case. `rigidBodyAndBeam` became runnable and benchmark outputs were committed (`a94db8d`, `ab56a8d`).

5-7 MAY
Diagnostic and authoritative state. Logged BlockEigen body motion, exposed full state, copied it to moorFV, and zeroed the old restraint return to prevent double counting (`edfa305`, `c78419b`).

24 JULY - 11 AUGUST
Attachment audit. Added attachment-force history, explicit ownership modes, reciprocal terms and the Newmark/mass scale $\Delta t^2\beta/m \times CQW/L$. Staged and fully implicit experiments exposed a mismatch between beam increments and total body displacement.

30 SEPTEMBER
Conceptual reset. Documentation was corrected: the staged authoritative path was useful, but did not yet contain the final body residual and two-way Jacobian. A phased true-monolithic plan was adopted (`f84a03a`, `4ab6dc8`).

Why the earlier path was not the final monolithic solver

- ! The appended body block remained effectively an identity update in staged mode.
- ! Fully implicit experiments mixed an incremental beam equation with a total body displacement quantity.
- ! Velocity/acceleration could still originate from explicit force feedback rather than the converged displacement.
- ! `FvBeamNewmark` could recompute the body after the BlockEigen solve.

> **Value retained.** These experiments established state transfer, Newmark terms, force diagnostics, sign conventions, ownership checks and regression infrastructure used by the final implementation.

Evidence: `documentation/{BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md, BEAM_ATTACHMENT_KINEMATICS_OWNERSHIP.md, rigidBodyAndBeamCoupledSolverDevelopmentLog.txt, rigidBodyImplicitAttachmentCoefficientChanges.txt}`.

DEVELOPMENT HISTORY II

True monolithic phases 0-2

Phase	Implementation	Verification outcome
0 `a2dcb75`	Beam-owned `rigidBodyEnd`; one switch `none | partitioned | monolithic`; fast no-fluid tests.	7 analytic checks pass in about 25 s. Hanging load/stretch, exact axial mode and pendulum period verified. Repeated-predictor and gravity/rho initialisation bugs fixed.
1 `359e651`	Translation residual in force units; attachment eliminated into body columns; state recovered from converged displacement; FD check.	24 checks pass. Body columns match finite differences to 1e-9 on significant entries. Monolithic differs from converged partitioned by at most 3e-5 of peak and uses about one-third of beam Newton iterations.
2 `03bbd7d`	Finite rotation, global inertia, gyro and damping, r x F, geometric terms, SO(3) Jacobian.	54 checks pass. Current worst FD column error is 6.25e-6 (tolerance 1e-5); cross-method differences remain 1e-4 of peak (5e-4 in compound pendulum); 2.3-4.4x fewer Newton iterations.

Hanging body

Body swings 90 degrees and settles. Centre-of-mass vertical misalignment: **1.6e-6 rad**. Force within **0.44%** of mg.

Free rotation

Asymmetric torque-free body. Global angular momentum conserved to **1e-6**; kinetic energy to **3e-8**.

Compound pendulum

Both linearised rod-body modes recovered. Frequency errors: **8e-4** and **4e-3**.

Representative Newton behaviour

A translation test reduced its residual approximately `7.4e3 → 1.5e−2 → 1.3e−5`, consistent with near-quadratic convergence once close to the solution.

> **Acceptance criterion met.** Analytic correctness, a finite-difference Jacobian, converged partitioned agreement and lower nonlinear work all point to the same implementation, rather than relying on a single benchmark.

Evidence: `tutorials/rigidBodyAndBeam/rigidBodyEnd/{README.md,checkRigidBodyEnd.py}`; commit messages for `a2dcb75`, `359e651`, `03bbd7d`.

TEST INVENTORY

Standalone analytic and conservation suite

These tests isolate beam-body mathematics from fluid coupling and run quickly enough to serve as the main regression suite.

Case	What is exercised	Independent check
`hangingMass`	Static axial load and beam/body equilibrium.	End force = -mg; extension includes end mass and half rod weight.
`axialOscillation`	Undamped translation/Newmark dynamics.	Exact rod-plus-end-mass root `z tan z = m_rod/m_body`; mean and peak displacement.
`pendulum`	Gravity, transverse deformation and body translation.	Finite-amplitude period with stretch and bending boundary-layer correction.
`hangingBody`	Large rotation, offset centre of mass, damping.	Beam force and final vertical alignment.
`freeRotation`	3-D inertia, gyroscopic term, orientation update.	Angular momentum and energy conservation.
`compoundPendulum`	Coupled beam and rotating-body modes.	Both eigenfrequencies of the linearised double-pendulum model.

Checks applied to every eligible case

Analytic

Period, equilibrium, force, extension, conserved quantity or mode frequency.

Differential

Monolithic body columns versus central finite-difference residual perturbations.

Cross-method

Monolithic history versus partitioned coupling iterated to a tight tolerance.

Physical caveat in compoundPendulum. The initially unstretched rod carries an undamped axial mode, causing real energy exchange and beats; the modal frequencies remain correct. A thinner, stiffer alternative risked loss of tension and buckling.

The suite's `Allrun` duplicates each base case with `rigidBodyCoupling monolithic`, runs both, then executes `checkRigidBodyEnd.py`. A failed numerical or analytic check returns a non-zero status.

Evidence: `tutorials/rigidBodyAndBeam/rigidBodyEnd/README.md` and `checkRigidBodyEnd.py`.

MOORFV INTEGRATION

Phases 3-4: from line-body solve to fluid-loaded cases

Phase 3 - authoritative monolithic state

moorFV
Fluid force, weight, other restraints and constraints

→

finiteVolumeBeam
Initialise body, pass external load, call beam evolve

→

beamFoamCoupled
Copy accepted x, Q, v, a, angular state without recomputation

56
current write times checked

2.2e-10
maximum current state-copy difference

358.7 s
current one-corrector runtime for either method

The current trajectory stays within 2.51e-4 of peak displacement of the loop case. The original Phase-3 check covered 25 write times; current outputs extend the check to 56. This verifies that moorFV stores the state solved by beamFoam, which the earlier staged path did not guarantee.

Phase 4 - PIMPLE outer-corrector study

Base-case method	Outer correctors	Max displacement error	Max force error	Interpretation
Partitioned loop	1	5.2e-4	2.0e-2	Beam/body lag dominates force.
Monolithic	1	2.7e-4	5.4e-4	Beam/body splitting largely removed.
Monolithic	8	~2e-6 versus converged partitioned loop		Outer-corrector convergence agrees.

Cost interpretation. At equal outer-corrector count, the fluid solve dominates elapsed time. Monolithic mainly reduces beam Newton work and allows accurate results with fewer outer exchanges when beam/body splitting is the limiting error.

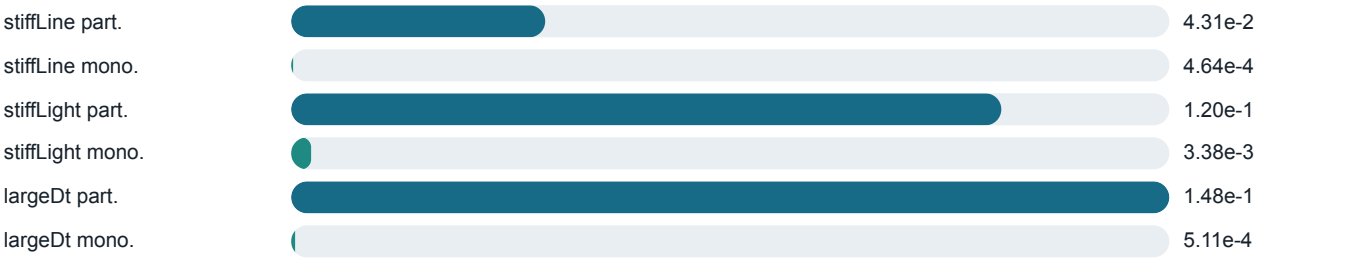
Evidence: commits `83acbbb`, `bf5bbbe`; `tutorials/rigidBodyAndBeam/rigidBodyAndBeam_beamFoamCoupled/checkCoupledState.py`; `documentation/source/monolithicCouplingPlan.html`.

STRESS VALIDATION

Short interFoam stress matrix

Each 0.25 s variant was run with the partitioned loop and monolithic solver at 1 and 8 PIMPLE outer correctors. Errors below are maximum force differences relative to the monolithic 8-corrector peak.

Variant	Change from base	Partitioned, 1 outer	Monolithic, 1 outer	Ratio
stiffLine	E x 100	4.31e-2	4.64e-4	93x lower
stiffLight	E x 100; mass / 10	1.20e-1	3.38e-3	36x lower
stiffLargeDt	E x 100; maxCo 1	1.48e-1	5.11e-4	290x lower
lightBody	mass / 10	7.19e-3	6.87e-3	similar



Beam Newton work. Monolithic used 18-37% fewer iterations with one corrector and 35-38% fewer with eight.

Limit exposed. In `lightBody` both methods are similarly inaccurate because the remaining error is explicit fluid/body coupling, not beam/body coupling.

No run diverged within 0.25 s; the one-pass partitioned force error grew by 15-115x over each stiff run. The study therefore demonstrates error growth and cost, not a long-time crash.

Evidence: `tutorials/rigidBodyAndBeam/rigidBodyAndBeam_stress/Allrun.out` and `summarise.py`; commit `bf5bbbe`.

ROBUSTNESS: PINNED BODY

Whirling end body - moment sweep

A 2 m rod carries a 1 kg body pinned 0.1 m off its centre of mass. A constant axial moment spins the body; centrifugal force whirls the rod. The base critical speed is about 10 rad/s. Both methods use the same time step and mechanics.

Setup: radius 0.05 m, E=5.4326e7 Pa, G=E/2.6, rod rho=1 kg/m3, 40 segments; body J=(0.05,0.01,0.01) kg m2, angular damping 0.03 N m s; no gravity. dt=0.001 s to 15 s, beta=0.25, gamma=0.5. Partitioned cap=50, tolerance=1e-10.

M (N m)	Final omega / omega_c	Partitioned Newton/step	Monolithic Newton/step	Runtime part./mono. (s)	Path difference / radius
0.075	0.250	6.81	3.00	31.5 / 14.4	6.8e-5
0.150	0.497	8.05	3.77	37.1 / 18.0	1.8e-5
0.225	0.717	8.74	3.90	40.3 / 18.5	5.2e-6
0.270	0.803	9.25	3.91	42.6 / 18.7	1.5e-6
0.300	0.841	9.48	3.93	43.9 / 18.9	4.2e-6
0.450	0.924	9.49	3.91	44.3 / 18.9	1.6e-5
0.900	1.036	9.59	3.82	45.6 / 18.8	1.2e-4

7/7
moments finished with both methods

2.2-2.4x
monolithic runtime speed-up

≤5.3%
linear-theory radius error where applicable

Physics captured. Above roughly 0.8 omega_c, spin no longer approaches M/c: energy feeds the whirl and spin stalls near critical speed (Sommerfeld effect). At 0.9 N m, the tip radius reaches 0.82 m. Both methods handle the large motion.

This moment sweep stresses large rotations and deformation more than coupling stability; the stiffness sweep provides the sharper partitioned/monolithic separation.

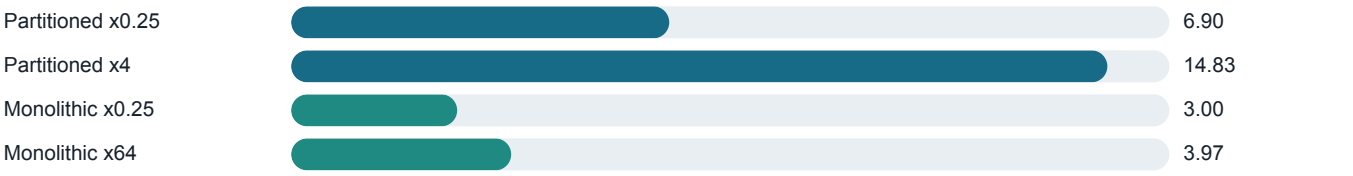
Committed evidence: `tutorials/rigidBodyEndWhirl/{README.md,summary.txt}`, commit `c010838`.

ROBUSTNESS: PINNED BODY

Whirling end body - stiffness sweep

The relevant fixed-point stability measure is approximately $S = \beta \Delta t^2 EA / (L\, m_{pin})$. As S increases, the partitioned exchange needs more iterations and ultimately grows instead of contracting.

EI factor	S	Part. coupling mean (max)	Part. Newton/step	Mono. Newton/step	Status
0.25	0.027	2.21 (4)	6.90	3.00	both finish
0.5	0.053	2.46 (5)	7.17	3.00	both finish
0.75	0.080	3.31 (6)	8.41	3.86	both finish
1	0.107	3.46 (6)	8.74	3.90	both finish
2	0.213	3.79 (8)	9.41	3.92	both finish
4	0.427	6.50 (14)	14.83	3.94	both finish
16	1.707	50 (cap)	-	3.96	PART. FAILS T=.002
64	6.827	diverges immediately	-	3.97	PART. FAILS T=.001



Strongest standalone result. Monolithic cost is essentially stiffness-independent (3.0-4.0 Newton iterations/step) and completes every case. Partitioned cost more than doubles before failing between S=0.427 and 1.707.

Where both finish, paths differ by at most 4.7e-4 of the tip radius. Monolithic final radii remain within 1.5% of linear theory where that check is valid.

Committed evidence: `tutorials/rigidBodyEndWhirl/summary.txt`.

NEWLY COMMITTED CAPABILITY

Welded end: twist and direct moment transfer

> **Status:** implementation committed in `99f35f7`; twist sweeps, moorFV case and comparison plots committed in `ffa7ca3` on 12 October. Results below describe those local tests.

Code extension

- `joint pin | welded`; welded requires body inertia and a fixed-value Theta boundary.
- Partitioned mode turns the beam end with the body and feeds both `-Q` and `-M` back.
- Monolithic mode adds end-face rotation columns, beam moment residuals and derivatives to the same BlockEigen solve.
- Finite-difference `jacobianCheck` covers the new terms and is reported to agree to round-off.

Standalone twist results

Setup: same 2 m, 40-segment base rod; body mass 1 kg, J=0.05 kg m2 at the tip, angular damping 1 N m s. dt=0.001 s to 1.5 s; beta=0.25, gamma=0.5; coupling cap=50, tolerance=1e-10. moorFV torsion uses dt=0.002 s and three outer correctors.

Study	Partitioned	Monolithic	Agreement / theory
Static twist 0.1-4 rad	~4.6 Newton/step; 2.3 coupling passes	2.0 Newton/step	Twist error ~3e-7; moment error ~3e-6; histories within 5e-7 rad.
Stiffness G x0.25 to x1024	Coupling mean rises 2.15 to 28.15; Newton/step 4.23 to 56.66.	2.0 Newton/step throughout.	Where both finish, method difference ≤7e-7 rad.
G x4096, x16384	Fails at t=.002 and .003 s.	Finishes 1.5 s.	Large omega Δt leaves unresolved physical decay; final static errors 0.6% and 14.5% are time-resolution effects.
Twist 6 rad	Buckles/crashes at 1.328 s.	Later buckles if continued.	Above Greenhill torque; does not distinguish algorithms.

moorFV free torsion, 0.5 s

Against analytic amplitude: monolithic **0.9998** with maximum relative theory error **9.72e-3**; partitioned **1.3796** with error **0.384**. Runtime was 169.6 s versus 175.3 s. The largest twist-history difference was 0.383 of amplitude.

Evidence: `tutorials/rigidBodyEndTwist/{README.md,summary.txt,moorFV/comparison.txt}`; commits `99f35f7` and `ffa7ca3`.

LONG CFD COMPARISON

Floating box in waves: 10 s completed pair

A 0.1 x 0.05 x 0.05 m box is moored by one flexible beamFoam line in a wave tank. Both cases use the same mesh, line, gravity, seabed contact, constraints and three PIMPLE outer correctors. The only subsystem-coupling change is `FvBeamNewmark` versus `beamFoamCoupled`.

Quantity at 10 s	Monolithic	Partitioned	Partitioned - monolithic
Surge	-87.393 mm	-88.516 mm	-1.123 mm
Heave	+6.392 mm	+6.460 mm	+0.068 mm
Pitch	+2.756 deg	+2.770 deg	+0.014 deg
Axial force at box	+0.030 mN	-0.016 mN	-0.046 mN
Axial force at anchor	-0.528 mN	-0.578 mN	-0.050 mN

5008
time steps in both cases

~3835 s
execution time for each method

3.7%
difference in peak box tension

History-level comparison

- Maximum differences: 1.12 mm surge, 0.075 mm heave, 0.154 deg pitch.
- Peak box tension: 0.1044 N at 2.312 s (monolithic), 0.1006 N at 2.316 s (partitioned), 0.1098 N in MoorDyn.
- Free-surface probes differ by at most 0.11 mm and 0.03 mm.
- Mean summed beam Newton iterations/step: 12.72 monolithic, 12.08 partitioned. Wall and execution time are effectively equal because fluid work dominates and monolithic iterations are cheaper.

> **Finding.** The new solver reproduces the long partitioned CFD trajectory closely and does not impose an elapsed-time penalty in this case.

Committed documentation and run outputs: `tutorials/floatRigidBody/floatRigidBody_monolithic/README.md` (run 9 October 2026) and paired case logs.

FLOATING-CASE LESSONS

Constraints, placement, gravity, contact and diagnostics

Issue found	Observed effect	Correction / evidence
Missing `refWf` in attachment point	A rotated/translated line put the moment arm about 0.3 m wrong, producing spurious roll and yaw.	Attachment changed to `Cf + refWf + W` in `542d3c7`.
No body constraints in monolithic block	The half-submerged box is hydrostatically unstable in roll (GM = -0.0042 m); roll grew from round-off and the case diverged near 1.07 s.	Projection `P A P + C` for translation/rotation rows and columns; standalone 54 checks remained passing.
Line-region gravity not copied	Earlier coupled runs had weight in beam initialisation but a weightless line during interFoam.	`makeBeams` now copies `constant/g`; log confirms gravity source.
Ground-contact implementation	Wrong sign, missing cell-length scale and missing Jacobian caused divergence at first floor contact.	Fixed 8-9 October; normal stiffness is implicit, damping explicit. Up to 15 cells contact the floor after 2.60 s.
Force magnitude called “tension”	For a bent/slack line, most end force can be shear.	New axial-force diagnostic resolves Q along the line tangent; plots now distinguish axial tension and |Q|.

MoorDyn is a useful reference, not an identical line model

BeamFoam includes bending and can carry compression; MoorDyn's lumped-mass segments do not. BeamFoam uses submerged density because beam buoyancy is not active, while MoorDyn applies buoyancy. Seabed stiffness/damping also differ. Agreement in peak load is informative, but differences should not automatically be assigned to the coupling algorithm.

> **Restart limitation.** Partitioned cases can restart after copying missing reference fields. `beamFoamCoupled` currently requires the initial body state and rejects a displaced-state restart.

Evidence: commits `542d3c7`, `05c0b1b`, `20f037f`, `b8f641c`; floatRigidBody READMEs and logs.

CURRENT GENERATED OUTPUTS

20%-slack line: asymmetric completion result

The 12 October slack-line cases extend the unstretched line from 0.23585 m to about 0.28302 m. The beam line lies partly on the seabed and can push in compression; MoorDyn shortens its non-compressive line during initialisation. These are intentionally challenging, non-identical physical initial states.

Method	Status	Last time	Execution time	Beam Newton/step	Final surge / heave / pitch
Partitioned beamFoam	FAILED	12.442 s	4862 s	12.47	34.36 mm / 3.32 mm / -4.096 deg
Monolithic beamFoam	FINISHED	20.000 s	7806 s	13.19	7.20 mm / 1.00 mm / +3.176 deg
MoorDyn	FINISHED	20.000 s	7578 s	-	10.57 mm / 0.80 mm / +3.057 deg

Loads and common-history comparison

- Partitioned run fails inside `evolveBeam()` at the 200-iteration limit: 6,221 completed logged steps, then the failing step at 12.442 s.
- Monolithic peak axial force is 0.0291 N at the box and 0.0286 N at the anchor near 6.402 s; compression immediately after the event reaches about -0.0067 and -0.0072 N.
- Using the final row per time step, partitioned reaches 0.00763 N at the box and 0.00712 N at the anchor near 6.458 s. Its largest intermediate outer-corrector box force is 0.01170 N.
- Across the common motion history, maximum monolithic-partitioned differences are 5.22 mm displacement and 0.00264 rad rotation.

Interpret carefully. This is a current-output finding, not yet a committed summary. The copied slack-line README still contains the earlier 10 s tight-line comparison and should be updated. Because the partitioned run stops early, final-state accuracy cannot be compared at 20 s.

Robustness signal. With the same beam initialisation, the monolithic case crosses the strong load/compression event and completes; the partitioned case does not.

Current logs: `tutorials/floatRigidBody_slackLine/*/log.interFoam`, motion histories and `postProcessing/0/axialForcebeamone.dat`. Cases introduced by `f2fc775`.

RECENT PROGRESSION

October integration milestones

1 OCTOBER
Phase 3 moorFV solver interface and Phase 4 outer-corrector/stress validation completed. State-copy and stress results committed.

5 OCTOBER
Constraint projections and non-zero reference placement fixed. Floating-box monolithic cases became viable.

6-7 OCTOBER
Base and 8-corrector float comparisons added; missing line gravity found; MoorDyn reference and axial-tension diagnostic added.

8-9 OCTOBER
Ground-contact sign, scale and tangent fixed; region-aware function objects added. Paired 10 s monolithic/partitioned wave runs completed.

12 OCTOBER
Rotation tensors made writable to retain placed/curved initial states; 20%-slack line cases added. Pinned whirl robustness sweeps committed.

12 OCTOBER, EVENING
Welded end and moment transfer committed in `99f35f7`; standalone and moorFV twist tests committed in `ffa7ca3`.

Implementation maturity gained through testing

Correctness controls

- ✓ Analytic checks
- ✓ Central FD Jacobian
- ✓ Converged partitioned reference
- ✓ State ownership regression
- ✓ Constraint regression

Operational diagnostics

- ✓ Body and attachment histories
- ✓ Axial tension versus |Q|
- ✓ Newton and coupling iteration counts
- ✓ Runtime and time-step histories
- ✓ Contact-cell reporting

Archival trials and incomplete early runs

The July/August staged short case completed to 0.01 s. A full case initially failed at 0.02 s; after the stale boundary timing fix it passed that point to 0.07 s and was stopped manually. The reciprocal-coefficient note records compile verification only.

Floating-case trial folders include setup-only copies. The gravity pair completed 10 s; archival weightless runs reported a 13.04 mm final surge gap at three correctors, reduced to about 1.3 mm at eight. These precede the final gravity/contact setup. Folder existence alone is not evidence of a completed test.

Commit chronology: `83acbbb`, `bf5bbbe`, `542d3c7`, `6b295d9`, `05c0b1b`, `20f037f`, `b8f641c`, `f2fc775`, `c010838`.

SYNTHESIS

What the tests establish

Claim	Evidence	Confidence
Same converged mechanics	Analytic cases and tight partitioned references agree within 1e-4 of peak, or 5e-4 in compoundPendulum, in the committed pin suite.	HIGH
Correct coupled derivatives	Current FD checks remain below 1e-5 (worst 6.25e-6); translation-only cases reach 5.15e-10. Welded terms are reported at round-off.	HIGH
Better stiff-coupling robustness	Whirl and twist stiffness sweeps show flat monolithic iteration counts while partitioned exchange slows and then diverges.	HIGH
moorFV state is truly authoritative	BeamFoam-solved and moorFV-stored states agree to ~2.2e-10 at all 56 current checked write times.	HIGH
Works in a realistic CFD case	Paired 10 s wave runs, 5008 steps, close motion/load agreement and equal elapsed cost.	HIGH
Eliminates all FSI instability	False: light-body stress case remains limited by explicit fluid load; PIMPLE correctors are still needed.	NOT CLAIMED
General multi-line/parallel solver	Only one finiteVolumeBeam restraint and single-face attachment are supported in demonstrated code.	NOT YET

Overall conclusion

For a single attached rigid body and one beam line, the true monolithic formulation has layered evidence for correctness, materially improves stiff-coupling robustness, and is integrated into the intended moorFV workflow. The next work is generalisation, restart/parallel support, routine welded regression coverage, and consolidation of slack-line results.

Ready for continued benchmark use: committed pin-joint standalone and single-line moorFV cases.

Needs further consolidation: automated welded regression coverage and slack-line result documentation.

Synthesis uses all evidence listed in the source index on page 21.

REMAINING WORK

Recommended validation and engineering priorities

Priority	Action	Acceptance evidence
1	Make welded coupling a routine regression	Add twist suite to routine checks; retain FD Jacobian; document the moorFV partitioned amplitude error and distinguish it from time-integration error.
2	Close the slack-line comparison	Update copied READMEs; retain failed partitioned log; rerun with controlled time step/coupling settings to locate the stability boundary without hiding the failure.
3	Automate CI-sized coverage	Short pin translation, rotation, welded torsion, invalid configuration, constraint and state-copy tests with numeric thresholds.
4	Implement multiple lines	Static condensation or an assembled multi-line tangent; compare four-line motion/load with a converged partitioned reference.
5	Parallel and restart support	Attachment patch ownership across processors; write/read body and beam reference/dynamic states; restart from a displaced body.
6	Clarify convergence norm	Review body-state contribution to the X-norm denominator; show tolerance invariance on a stiff case.
7	Separate fluid/body limitations	Added-mass sweep over body/fluid density, outer correctors and acceleration treatment; report stability independently from beam/body coupling.

Known constraints to preserve in user documentation

- ! Exactly one `finiteVolumeBeam` restraint in `beamFoamCoupled`.
- ! Centre of rotation must equal centre of mass for the current moorFV monolithic path.
- ! Attachment patch must resolve to one face on the participating processor.
- ! Other restraints and fluid force are external loads for each beam-body solve.
- ! Displaced-state restart is rejected; reference/dynamic beam fields are not all written.
- ! High-frequency twist cases need a time-step study even when the nonlinear solve converges.

> **Recommended release message.** “Monolithic beam-attached rigid-body coupling for one line, with pin support validated and welded support experimental; fluid coupling remains partitioned through PIMPLE.”

Code limits: `beamFoamCoupled.C:76–110`; `finiteVolumeBeam.C:571–587`; `coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C:180–185`.

APPENDIX

Evidence and reproducibility index

Core implementation

- `src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/coupledTotalLagNewtonRaphsonBeamRigidBodyEnd.C`
- `src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/rigidBodyEnd/{rigidBodyEnd.C,rigidBodyEnd.H}`
- `src/wireBunchingModels/beamModels/coupledTotalLagNewtonRaphsonBeam/RigidBodyCouplingData.H`
- `src/wireBunchingModels/numerics/BlockEigenSolvers/BlockEigenSolverOF/{BlockEigenSolverOF.C,BlockEigenSolverOF.H}`
- `../sDoFRGBFvBeam/sixDoFFvBeamSolvers/{FvBeamNewmark,beamFoamCoupled}` and `.../finiteVolumeBeamRestraint`

Tests and outputs

- `tutorials/rigidBodyAndBeam/rigidBodyEnd` - analytic/conservation/Jacobian suite.
- `tutorials/rigidBodyAndBeam/rigidBodyAndBeam_stress` - 1/8-corrector stress matrix.
- `tutorials/rigidBodyEndWhirl` - committed pin-joint moment and stiffness sweeps.
- `tutorials/rigidBodyEndTwist` - committed welded-joint sweeps and moorFV comparison.
- `tutorials/floatRigidBody` - completed 10 s beamFoam/MoorDyn wave cases.
- `tutorials/floatRigidBody_slackLine` - current 20 s/failed-12.442 s outputs.

Development records

- `documentation/source/monolithicCouplingPlan.html`
- `documentation/{BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md,BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md}`
- `documentation/{rigidBodyAndBeamCoupledSolverDevelopmentLog.txt,rigidBodyKinematicCouplingDevelopment.txt,rigidBodyImplicitAttachmentCoefficientChanges.txt}`

Normal rerun entry points

Purpose	Command/location
Build	`./Allwmake` from beamFoam root; then moorFV build for coupled integration.
Fast analytic suite	`tutorials/rigidBodyAndBeam/rigidBodyEnd/Allrun`
Pin robustness	`tutorials/rigidBodyEndWhirl/Allrun [moment|stiffness]`
Weld robustness	`tutorials/rigidBodyEndTwist/Allrun [moment|stiffness]`
Full wave case	Paired `Allrun` scripts under `tutorials/floatRigidBody`; approximately 65 min each in serial.

> **Report conclusion:** validated single-line monolithic coupling is technically credible and demonstrably robust; generalisation and consolidation are the remaining steps.

This report was generated from the local repository only. It contains no external web claims.